

SMART INDIA HACKATHON 2025



TITLE PAGE

- **Problem Statement ID – 25049**
- **Problem Statement Title-AI-Driven Public Health**

Chatbot for Disease Awareness

- **Theme- HealthTech**
- **PS Category- Software**
- **Team ID-**
- **Team Name - Tech type**



AI -BASED MEDICINE REMINDER AND HEALTH MONITORING

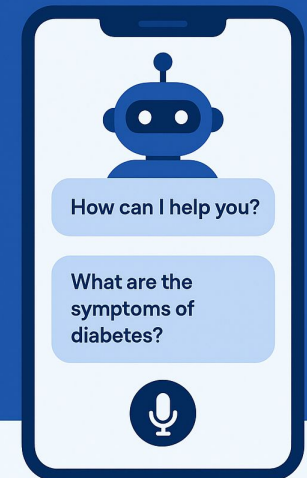
PROBLEM STATEMENT

- People lack easy access to reliable disease information.
- Misinformation leads to wrong self-diagnosis.
- Doctors are overloaded with basic health queries.
- Low awareness of preventive measures causes disease spread.

IDEA/SOLUTION

- ◆ AI-driven intelligent chatbot offering context-aware health guidance
- ◆ Bridges healthcare accessibility gaps in remote and rural regions
- ◆ Provides preventive care tips and trustworthy medical information
- ◆ Multi language support for best interaction in rural areas
- ◆ Sends timely notification

AI-DRIVEN PUBLIC HEALTH CHATBOT FOR DISEASE AWARENESS



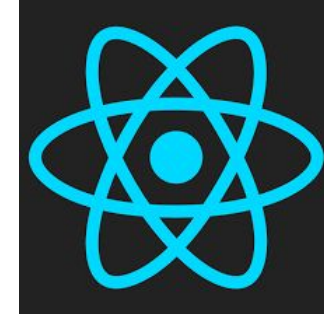
- ✓ Disease Awareness & Information
- ✓ Symptom Checker
- ✓ Preventive Health Tips
- ✓ 24/7 Availability



Technical Approach

◆ Technical Approach:

- Local Database (SQLite) for offline functionality
- Notification API for timely reminders
- Health Monitoring Module for user input
- Optional Cloud Sync for data backup



Technologies Used:

React.js) → Node.js & Express.js → TensorFlow PyTorch →
MongoDB / Firebase → Dialogflow / Rasa (NLP) →
Multi-language



Visibility

Identifying the Core Problem

- **Easy-to-use interface ensures high user adoption.**
- **Clear medicine schedules improve user compliance.**
- **Real-time notifications keep users informed.**

Feasibility

- **What: AI-based multilingual health chatbot via WhatsApp.**
- **Why: Improve health literacy, early disease detection, support ASHAs.**
- **Who: Rural & semi-urban population, health workers, local officials.**
- **Where: Pilot → UP scale → National integration.**
- **How: AI + human-in-loop → curated content → phased deployment.**
- **Resources: \$45K–\$90K initial, recurring costs for API, cloud, and staff**

Viability

- **Goal: Long-term, sustainable AI health assistant.**
- **Why: Reduces hospital burden, empowers users, supports gov initiatives.**
- **How: Revenue via WhatsApp API, hosting subscriptions, licensing, analytics.**
- **When: Continuous operation with updates and feedback**
- **ROI: Long-term cost savings in healthcare + potential revenue.**

Impact & Benefits of AI Health Awareness Chatbot

- Minimizes medication non-adherence and associated health risks.
- Reduces unnecessary hospital admissions and healthcare costs.
- Enhances patient well-being through timely interventions.

Benefit:

- Ensures seamless offline operation for remote areas.
- Provides automated, timely medicine notifications.
- Integrates simple health monitoring (BP, Sugar) for proactive care.



Research

- **Health Challenges in UP:**
 - Dual disease burden:
→ Non-communicable (obesity, deficiencies) + Infectious (Japanese Encephalitis, Dengue).
- **Health Literacy Issue:**
72.6% adolescents have only *moderate health literacy*
- **Need for Smart Communication**
- **Case Study – ASHABot (Rajasthan):**
 - AI + human-in-loop model



World Health Organization (WHO) – Health Literacy

<https://www.who.int/health-topics/health-literacy>

National Vector Borne Disease Control Programme

<https://nvbdcp.gov.in/index4.php>

ASHABot Project

<https://www.digitalgreen.org>

2. References

- India Adolescent Health Study → 72.6% moderate health literacy.
- ASHABot Project → AI + human support in rural health.
- Indian Public Health Manuals + Immunization Guidelines → Knowledge source.
- UP Health Data → Japanese Encephalitis, AES, Dengue outbreaks.